

Multiple Choice

1. **Answer:** A

Options: A) 9180 mg; B) 9150 mg; C) 9200 mg; D) 9135 mg; E) 9155 mg

Note: Correct option: A - 9180 mg

2. **Answer:** A

Options: A) 2.5×10^{-3} ; B) 5.5×10^{-3} ; C) 1.5×10^{-3} ; D) 1.0×10^{-3} ; E) 1.2×10^{-3}

Note: Correct option : A – 2.5×10^{-3}

2. **Answer:** A

Options: A) H_2PO_4^- , HPO_4^{2-} , H_3PO_4 ; B) HPO_4^{2-} , H_3PO_4 , H_2PO_4^- ; C) HPO_4^{2-} , H_2PO_4^- , H_3PO_4 ; D) H_2PO_4^- , H_3PO_4 , PO_4^{3-} ; E) PO_4^{3-} , HPO_4^{2-} , H_2PO_4^-

Note: Correct option : A – H_2PO_4^- , HPO_4^{2-} , H_3PO_4

3. **Answer:** D

Options: A) 9.20 M; B) 12.10 M; C) 5.60 M; D) 7.83 M; E) 2.95 M

Note: Correct option: D - 7.83 M

4. **Answer:** C

Options: A) The air feels cold because it mixes with colder atmospheric air; B) The air feels cold due to a chemical reaction within the tire; C) The air temperature falls because it is doing work and expending energy; D) The cooling sensation is a result of the air's rapid expansion; E) The cooling sensation is due to the air's humidity

Note: Correct option: C - The air temperature falls because it is doing work and expending energy

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '11.83'.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
10. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** 675 g. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.
12. **Answer:** 1.67 g. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key